

Michael Marques Goncalves

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EDUCATION

Embry-Riddle, College of Engineering. *Ph.D., Aerospace Engineering.* **GPA:** 4.0/4.0 May 2025

Embry-Riddle, College of Engineering. *M.S., Aerospace Engineering.* **GPA:** 3.9/4.0 July 2021

Universidade Federal de Minas Gerais, Brazil. *B.S., Aerospace Engineering.* December 2018

SKILLS

Technologies: Star-CCM+, OpenFOAM, CDI-CHARM, Paraview, OpenVSP, UCD-QuietFly, PSU-WOPWOP, Tecplot, MATLAB, XFOIL, Pointwise, CATIA, Simulink.

Programming Languages: Python, Bash, Fortran, Latex.

EXPERIENCE

Embry-Riddle Aeronautical University Daytona Beach, FL
Postdoctoral Research Scholar Aug 2025

- Conducting CFD analysis using Star-CCM+ for NASA Dragonfly rotorcraft mission, focusing on rotordynamics and dust mobilization in Titan (Saturn's moon).
- Lecturing four sections of Introduction to Aerospace Flight Vehicles class.
- Managing graduate students on the NASA ULI research project I was part of during my PhD.

Research Granted by NASA University Leadership Initiative Daytona Beach, FL
Graduate Research Assistant Aug 2023 - May 2025

- Built a rotorcraft aeroacoustic framework integrating BEMT with PSU-WOPWOP, using DES-based external flow simulations and BEMT-derived loading to assess aerodynamic performance and accurately predict noise sources, including ground-effect influences.
- Designed a broadband self-noise model using UCD-QuietFly, BEMT, XFOIL, and 2D airfoil CFD simulations to study noise trends across varying flight conditions.
- Created a converter tool linking OpenFOAM CFD output to PSU-WOPWOP, enabling reliable high-fidelity aeroacoustic analysis.

Joby Aviation Santa Cruz, CA
Aeroacoustics Intern June - Aug, 2024

- Contributed to broadband acoustic prediction tools by evaluating simulation methodologies, developing noise models, and running CHARM and UCD-QuietFly simulations. Results were validated with experimental measurements from NASA NFAC wind-tunnel facilities.
- Evaluated rotor blade aerodynamics using 2D RANS through NASA OVERFLOW and CHARM simulations, focusing on pressure distribution and unsteady wake interactions to support broadband noise modeling.
- Delivered weekly technical presentations and collaborated across teams, strengthening communication skills and technical reporting. Internship culminated in a paper presentation at AIAA SciTech 2025.

Research Granted by Office of Naval Research Daytona Beach, FL
Graduate Research Assistant January - July, 2023

- Performed DES of supersonic jet plume external flow to assess shock-cell structures and proposed a novel unsteady excitation strategy for jet noise reduction, demonstrating its effectiveness through detailed flow and acoustic analysis.
- Developed an in-house solver for the Ffowcs Williams-Hawkings equation using Farassat's 1A time-domain formulation for permeable surfaces; validated results against experimental jet noise data.

AWARDS & MILESTONES

- **International Engagement Award for Outstanding Teaching/Mentoring**
Embry-Riddle Aeronautical University April 2025
- **Treasurer**
Graduate Student Council - ERAU Jan 2025 - May 2025
- **Fundraising Chair**
Brazilian Student Association - ERAU May 2023 - May 2024
- **Graduate Commencement Student Speaker**
Embry-Riddle Aeronautical University May 2025
- **Participant**
Future Leaders in Aerospace Symposium May 2025

PUBLICATIONS

- Multi-Fidelity Assessment of eVTOL Propeller Noise in Ground-Effect Conditions. Michael Marques, Vladimir Golubev and Anastasios Lyrantzis. In *International Journal of Aeroacoustics*, February 2025. DOI: 10.1177/1475472X261419107
- Multi-Fidelity Assessment of eVTOL Propeller Noise in Ground-Effect Conditions. Michael Marques, Vladimir Golubev and Anastasios Lyrantzis. In *AIAA Aviation Forum*, July 2025. DOI: 10.2514/6.2025-3002
- Noise Reduction Using Synthetic Microjet Excitation in Supersonic Rectangular Jets. Michael Marques, Surabhi Singh, Anastasios Lyrantzis and Vladimir Golubev. In *Journal of Applied Sciences MPDI*, 2025. DOI: 10.3390/app15031180
- Evaluating Ground Effect on eVTOL Rotor Noise. Michael Marques, Vladimir Golubev, Anastasios Lyrantzis and Reda Mankbadi. In *AIAA SciTech Forum*, January 2025. DOI: 10.2514/6.2025-1242
- Assessment of Self-Noise Models Using NFAC Measurements of the Joby Aviation Propeller. Nikos Trembois, Michael Marques, Austin Thai and Jeremy Bain. In *AIAA SciTech Forum*, January 2025. DOI: 10.2514/6.2025-1045
- Acoustic Analysis of Noise Control of Rectangular Jet Using Micro-Jet Excitation. Michael Marques, Vladimir Golubev and Anastasios Lyrantzis. In *30th AIAA/CEAS Aeroacoustics Conference*, May 2024. DOI: 10.2514/6.2024-3308
- High-Fidelity Simulations of Vertiport Ground Effects on eVTOL Rotor Noise. Michael Marques, Vladimir Golubev, Anastasios Lyrantzis and Reda Mankbadi. In *30th AIAA/CEAS Aeroacoustics Conference*, May 2024. DOI: 10.2514/6.2024-3333
- Analysis of e-VTOL Rotor Noise Predictions. Michael Marques, Vladimir Golubev and Anastasios Lyrantzis. In *AIAA SciTech Forum*, January 2024. DOI: 10.2514/6.2024-2809
- High-Fidelity Simulations of Rectangular Jet Noise Control Using Micro-Jet Excitation. Michael Marques, Surabhi Singh, Sam Salehian, Vladimir Golubev and Anastasios Lyrantzis. In *AIAA SciTech Forum*, January 2024. DOI: 10.2514/6.2024-2306
- Comparative High-Fidelity Studies of Supersonic Rectangular and Round Jet Nozzle Flow. Michael Marques, Sam Salehian, Surabhi Singh, Vladimir Golubev, Anastasios Lyrantzis and Reda Mankbadi. In *AIAA Aviation Forum*, June 2023. DOI: 10.2514/6.2023-4517
- High Fidelity Simulations of Active Control of Coherent Structures in Axisymmetric Jet. Michael Marques, Vladimir Golubev, Sam Salehian and Reda Mankbadi. In *AIAA SciTech Forum*, January 2023. DOI: 10.2514/6.2023-0025
- Active Control of Coherent Structures in an Axisymmetric Jet. Michael Marques. *Master's Theses*, August 2021. <https://commons.erau.edu/edt/601/>